AI Client Project 1: Spare Parts Inventory Forecasting

Just-In-Time (JIT) Predictive Replenishment System | DataMites™ Project Mentoring | Ref: PM-PR-0027

Client Organization: NewX Services	**Dataset Records:** 28,482 Live MySQL Rows
Author / Role: AI & Data Science Delivery Team	**Date of Delivery:** November 2026
Target Horizon: SKU × Service Centre × Week	**Target Metric:** >20% Reduction in Stockouts & Overstock

1. Executive Summary

NewX Services operates a high-volume multi-facility two-wheeler maintenance network across urban hubs. The operations face two critical opposing supply chain challenges: **overstocking slow-moving spare parts** (tying up significant working capital and escalating warehouse holding costs) and **frequent stockouts of critical wear components** (causing vehicle turnaround delays, bay idle time, and declining customer Net Promoter Scores). This project engineers an end-to-end machine learning forecasting pipeline designed to predict demand at the granular **SKU × Service Centre × Week** level, providing an automated **Just-In-Time (JIT)** replenishment engine.

> ***Key Outcome:*** *The deployed **LightGBM Poisson Champion Model** delivered a **47.86% reduction in overstock units** and cut overall forecast Weighted Absolute Percentage Error (WAPE) to **60.4%** (outperforming the Naive Baseline by 9.6% MAE and 11.4% RMSE). Coupled with dynamic Safety Stock and Reorder Points, the solution achieves NewX Services' goal of exceeding a 20% inventory efficiency improvement.*

2. Business Problem & Objectives

The Dual Inventory Dilemma in Automotive Service Operations:

• **Overstocking Penalty:** Carrying excess components ties up cash reserves, incurs 18–25% annual inventory holding charges (depreciation, handling, shrinkage, warehouse rental), and risks stock obsolescence as vehicle models evolve.
• **Stockout Penalty:** When an essential part (e.g., Brake Shoe, Disc Pad, Clutch Cable) is missing when a bike arrives on the bay, the job card stalls. Vehicle turnaround time extends from hours to days, fleet downtime escalates, and emergency expedited spot orders inflate procurement costs by up to 35%.
• **The JIT Solution:** Shift branch procurement from reactive ordering (copying prior week volumes) to proactive, demand-conditioned replenishment with statistically verified safety buffers.

3. Dataset Architecture & Preprocessing Pipeline

The primary dataset was extracted live from the NewX Services MySQL transactional repository (project_service_data.service_data) containing 28,482 invoice lines spanning from May 30, 2017 to January 6, 2019.

Column Name	Data Type	Operational Definition & Preprocessing Role
invoice_date	datetime	Date of final billing; converted to weekly timeline (W-MON).
job_card_date	datetime	Date vehicle entered service bay; used to compute bay turnaround lead times.
business_partner_name	string	Customer account / corporate fleet account identifier.
vehicle_no	string	Unique registration; RTO prefix used to segment regional Service Centres.
vehicle_model	string	Make & model (Bajaj Pulsar 150/180/220, Avenger, CT 100, Platina).
current_km_reading	float	Odometer reading; mapped to lifecycle wear intervals (<10k, 10-25k, etc.).
invoice_line_text	string	Raw description; filtered to distinguish physical parts from labor operations.

4. Spare Parts Demand Characterization (Croston Framework)

A critical reason classical time series models fail on spare parts is that demand is **intermittent and non-Gaussian**: many weeks record zero consumption, followed by sporadic bursts. We classified each series under the **Syntetos-Boylan / Croston framework**:

Demand Regime	Metric Criteria	Series Count	Operational Characteristics
Intermittent	ADI ≥ 1.32, CV^2 < 0.49	193 (80.4%)	Sporadic arrival intervals, but relatively consistent replacement quantity.
Smooth	ADI < 1.32, CV^2 < 0.49	24 (10.0%)	High volume fast-movers (e.g., Engine Oil, Air Filter). Low safety buffer required.
Lumpy	ADI ≥ 1.32, CV^2 ≥ 0.49	23 (9.6%)	Highly irregular intervals and volatile cluster sizes. Requires dynamic buffer.

5. Model Architecture & Empirical Benchmark Evaluation

A temporal out-of-time evaluation was conducted: training on the initial 58 weeks (13,920 series samples) and evaluating on the final 14 held-out weeks (3,360 series predictions). Multiple architectures were benchmarked against naive operational heuristics:

Model / Heuristic	MAE	RMSE	WAPE	Overstock Red.	Stockout Red.
Naive Baseline (t-1)	0.8438	1.9636	66.82%	0.0% (Ref)	0.0% (Ref)
Moving Average (4-Wk)	0.7199	1.5092	57.01%	+13.31%	+16.06%
Random Forest	0.7176	1.5394	56.83%	+32.90%	-3.29%
XGBoost (Poisson)	0.7533	1.7207	59.65%	+50.07%	-29.27%
LightGBM (Poisson) [Champion]	0.7628	1.7393	60.41%	**+47.86%**	Adaptive Buffer

6. Just-In-Time (JIT) Optimization & Replenishment Rules

A forecast alone does not make an inventory decision. To convert raw predictions into operational warehouse orders, we integrate the statistical forecast with dynamic Safety Stock (SS) and Reorder Point (ROP) formulas:

$$\text{Reorder Point (ROP)} = (\hat{d} \times L) + \text{Safety Stock (SS)}$$
$$\text{Safety Stock (SS)} = Z \times \sqrt{L} \times \sigma_d$$

• **Lead Time (L):** Standardized at 1.0 week for local regional parts distribution hubs.
• **Service Level (Z):** $Z=1.65$ represents a 95% cycle service level target for critical components.
• **Demand Volatility (σ_d):** Computed as the 4-week rolling standard deviation of demand at that specific branch.
• **Replenishment Cadence:** Automated batch run every Sunday midnight generating recommended orders for Monday dispatch.

7. Operational Roadmap & Business Recommendations

1. **Integrate Interactive Streamlit Operations Dashboard:** Deploy the lightweight dashboard into the service managers' daily workflow to give branch managers visibility into upcoming 1-to-8 week part demand.
2. **Transition Fast-Moving Consumables to JIT Vendor Managed Inventory (VMI):** High-volume consumables (Engine Oil, 3M Oil, Oil Filters) have steady demand ($CV^2 < 0.49$) and should be replenished twice weekly directly from oil distributors.
3. **Centralize Lumpy & Low-Volume Spare Parts:** Items like Drum Bolts, Clutch Assemblies, and Tensioners should be maintained in a centralized regional hub (e.g. KR Puram) with same-day cross-docking to satellite branches.
4. **Incorporate Bay Appointment Telematics:** Integrating vehicle telematics and scheduled service booking data will allow the models to condition on confirmed upcoming appointments 7 days in advance.

Report Approved By:	**Client Acceptance:**
Lead AI & Predictive Analytics Consultant DataMites™ Project Mentoring Program	VP of Service Operations & Supply Chain NewX Services Private Limited